

Cartesian Coordinates

dx, dy, dz are infinitesimal displacements along X, Y and Z resp.

Differential quantities:

Length Element

$$d\vec{l} = \hat{x}dl_x + \hat{y}dl_y + \hat{z}dl_z$$

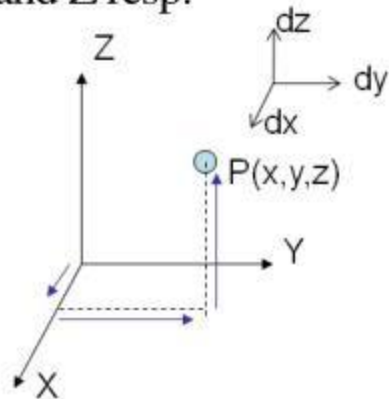
$$\text{or, } d\vec{l} = \hat{x}dx + \hat{y}dy + \hat{z}dz$$

Area Element

$$ds_x = dydz \quad ds_y = dxdz$$

$$ds_z = dxdy$$

$$d\vec{s} = \hat{x}ds_x + \hat{y}ds_y + \hat{z}ds_z \\ = \hat{x}dydz + \hat{y}dxdz + \hat{z}dxdy$$

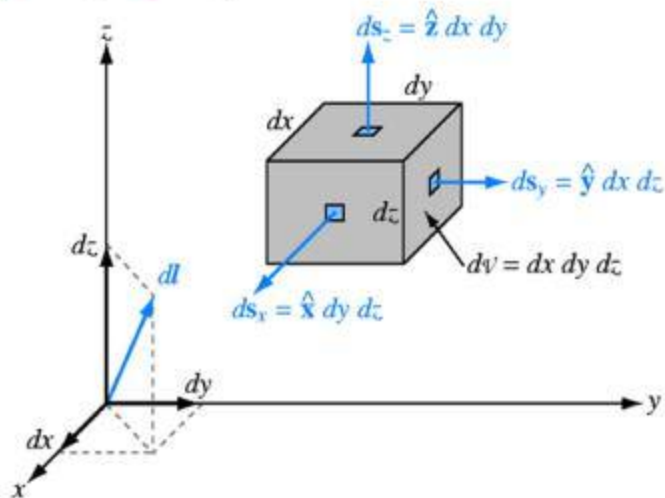


Volume:

$$dv = dxdydz$$

Ex: Show that volume of a cube of edge a is a^3 .

$$V = \int_v dv = \int_0^a dx \int_0^a dy \int_0^a dz = a^3$$



Spherical Coordinates

Differential quantities:

Length Element

$$dl_r = dr$$

$$dl_\theta = r d\theta$$

$$dl_\phi = r \sin \theta d\phi$$

$$\underline{d\vec{l} = \hat{r}dl_r + \hat{\theta}dl_\theta + \hat{\phi}dl_\phi}$$

Area Element:

$$ds_r = dl_\theta dl_\phi = r^2 \sin \theta d\theta d\phi$$

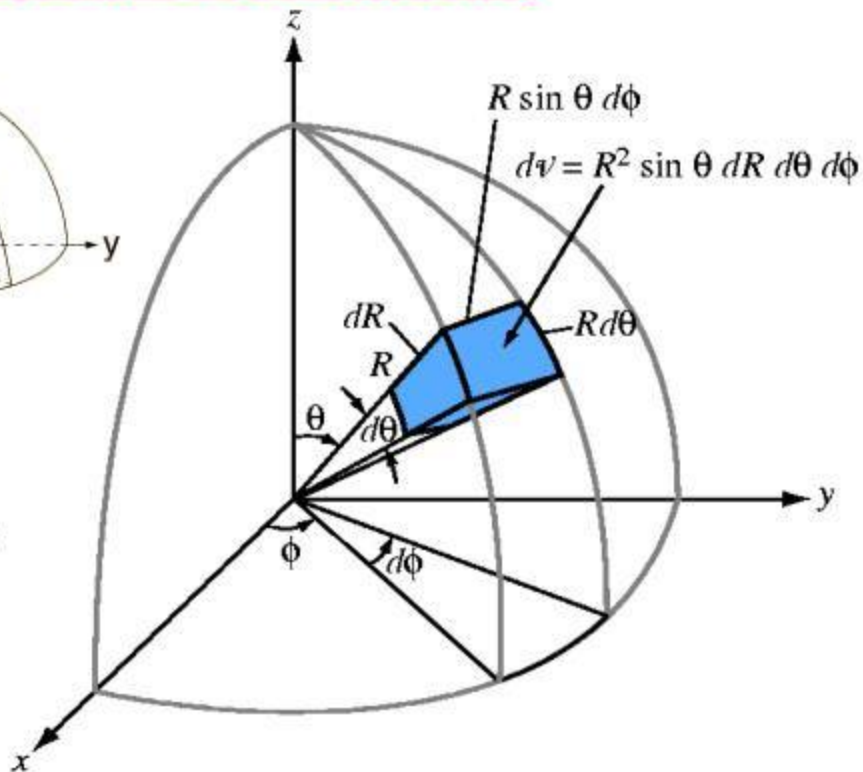
$$ds_\theta = dl_r dl_\phi = r \sin \theta dr d\phi$$

$$ds_\phi = dl_r dl_\theta = r dr d\theta$$

$$\underline{d\vec{s} = \hat{r}ds_r + \hat{\theta}ds_\theta + \hat{\phi}ds_\phi}$$

Volume Element

$$dv = r^2 \sin \theta dr d\theta d\phi$$



Find the volume and surface area of a sphere of radius R.